PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Prehospital risk assessment in patients suspected of Non-ST-segment elevation Acute Coronary Syndrome – A systematic review and meta-analysis
AUTHORS	Demandt, Jesse; Zelis, Jo; Koks, Arjan; Smits, Geert; van der Harst, Pim; Tonino, Pim; Dekker, Lukas; van het Veer, Marcel; Vlaar, Pieter-Jan

VERSION 1 – REVIEW

REVIEWER	Alotaibi, Ahmed The University of Manchester, Division of Cardiovascular Sciences
REVIEW RETURNED	23-Sep-2021

GENERAL COMMENTS	1- Please add the data source used and the date of the search in the abstract. 2- Include the search date in the methodology section. 3- In Quality Assessment Tool for Diagnostic Accuracy Studies (QUADAS) Tool table, please include what the letter (U) represents? I suppose it means unclear. 4- Figure 1: the words are not clear due to the small font size and font color. Try to increase the font size or change the font color to clarify it.

REVIEWER	Boogers, Mark Leiden Universitair Medisch Centrum, Cardiology
REVIEW RETURNED	05-Oct-2021

GENERAL COMMENTS	Review: “Prehospital Risk Assessment in Suspected Acute Coronary Syndrome Patients – A systematic Review and Meta-analysis” The review has evaluated the current literature regarding several patient parameters and diagnostics for risk stratification in prehospital setting. Combined risk scores as well as extramural assessment of biomarkers were predictors for identification of patients with a NSTE-ACS. Despite, these positive findings, I have some issues to address: 1. The objective of the systematic review could be more clearly described (using for instance the PICO system – Which exact population, Intervention as well as Outcome of the study was studied). 2. The current review of the literature for identification of best discriminator for prehospital triage in suspected ACS patient is bases upon different studies, predominantly non-randomized trials. In such an approach, the method of selection of included studies is truly important. In this perspective, I was wondering whether both reviewers were independently from each other selecting the included

	studied. In other words, was the search performed twice or only 1 time, and then checked by the second observer? 3. The same holds true for the data extraction and statistical analyses of the included studies. Was this done by 1 person? And rechecked by another? 4. There is in EMS studies always heterogeneity, what can influence the results of the systematic review. How did you address these study heterogeneity, can you explain in more detail? Especially, the study included both GP and EMS patient populations (patient heterogeneity) as well as different study endpoints (some had 3-day MACE rate, while others had final hospital diagnosis such as ACS or NSTEM/UAP). The authors should discuss the potential effects of heterogeneity in enrolled studies on their findings 5. Furthermore, why were studies with specific patient populations and small studies with less than 100 samples excluded? Some prehospital parameters can in such way be biased; for instance in STEMI patients, other ECG parameters are probably more relevant in STEMI patients than in patients with NSTEMI or thoracic chest pain without troponin release. 6. Can the authors be more detailed about the extracted data regarding ECG parameters (An “ischemic ECG” is rather a broad description, what is meant with this – ST depression, Negative T waves, ST segment elevation) and biomarkers (which specific generation or essay was used) ? 7. One of the limitation of EMS studies is often the lack of well-described pre-hospital patient and ECG data. For this reason, the lack of power of these prehospital parameters may be partly explained by lack of description. This should be discussed in more detail in the paper 8. The study inclusion is described in Figure 1, in which initially ~ 4000 studies were identified, and at the end only 15 studies were enrolled for the systematic review. The authors mention that all articles were screened by the primary author, and if suitable the were included. What is meant with this? Please describe what criteria were used for selection and check suitability. Additionally, can the authors explain in more detail what ‘’the removal of records meant’” as well as the ‘’exclusion of records’ in step 2 of this process? 9. What was the main exclusion criterium of all these studies? Please specify the exclusion reasons 10. The authors found that combined risk scores and biomarkers were better predictors in the prehospital scene than conventional patient, physical examination or ECG parameters. Any explanation for this finding (is the sum of parameters more reliable, than a single-parameter)? The value of combined risk scores depends highly on the medical experience, setting and sort of score you used. In general, the (pre) HEART scores are more often used at emergency departments or EMS, whereas the Marburg score is used in primary care setting. One should be cautious about drawing conclusions of the value of these combined scores in a particular setting.

REVIEWER	Sjoberg, Folke Linkopings universitet
REVIEW RETURNED	08-Jan-2022

GENERAL COMMENTS	This is a systematic review and meta analysis with the aim to review and compare available tools that have the best performance for pre-hospital risk assessment in suspected ACS patients. It is concluded that in the prehospital setting several diagnostic tools have been

	reported which could improve risk stratification, triage and, early treatment in suspected ACS patients. On-site assessment of Troponin and combined risk scores derived from the HEART score were the strongest predictors. These results support further studies to investigate the impact of these new tools on logistics and clinical outcome. This is a well thought of and undertaken systemic review on an important topic, that of identifying patients with ACS presenting in the prehospital or primary care setting. My main concern with this paper is that the review deals with only NSTE-ACS and this needs to be stated more clearly in the text throughout the manuscript. Also, emphasis needs to be made on the rather limited group of patients that constitutes the NSTE-ACS group. This can be further elaborated on in the discussion. When constructing risk stratification models it is a shortcoming that a part of the patient population, in this case the STEMI are excluded on beforehand. This could be even further stressed in study limitations.

VERSION 1 – AUTHOR RESPONSE

Responses to Reviewer #1

We thank the reviewer for his comments to improve our manuscript.

1. Please add the data source used and the date of the search in the abstract.
2. Include the search date in the methodology section.

Thank you for your suggestion. We included the data source and the date in the methods section of our abstract and in the main text.

“MEDLINE and EMBASE were searched up till April 1, 2021.” (Page 2)
“We performed searches in MEDLINE and EMBASE for all published articles up till April 1, 2021.” (Page 7)

Also described on page 2 and 7 in the methods sections, it states that all manuscripts before 1995 were excluded in the analysis.

3. In Quality Assessment Tool for Diagnostic Accuracy Studies (QUADAS) Tool table, please include what the letter (U) represents? I suppose it means unclear.

We thank the reviewer for his attentiveness. We have added the meaning of the letter “U” to page 5 of the Supplemental material.

“Y = yes; N = no; U = unclear” (Page 5)

4. Figure 1: the words are not clear due to the small font size and font color. Try to increase the font size or change the font color to clarify it.

Thank you for your suggestion. We increased the font size and hope it is easier to read now.

Responses to Reviewer #2

We thank the reviewer for his comments and suggestions to improve our manuscript.

1. The objective of the systematic review could be more clearly described (using for instance the PICO system – Which exact population, Intervention as well as Outcome of the study was studied).

We thank the reviewer for his comments. We agree that we could have been clearer about what exact population we were interested in. Therefore, we have changed the title of the manuscript and the following sentence in the manuscript.

“We performed a systematic review and meta-analysis of available prospective data to summarize and compare current available and new diagnostic tools for prehospital risk assessment in patients suspected of NSTE-ACS.” (Page 6)

“The objective of this review was to investigate which patient characteristics and diagnostic tools were used for early risk assessment in an undifferentiated prehospital population suspected of NSTE-ACS…” (Page 7)

The intervention we are interested is described on page 7 as well.
“…to investigate which patient characteristics and diagnostic tools were used for early risk assessment…”
Our outcome is described under the heading of “End Points and Definitions” on page 8.
“The primary end point was the final hospital discharge diagnosis of NSTE-ACS by the treating physician or Major Adverse Cardiac Event (MACE) within 6 weeks.”

2. The current review of the literature for identification of best discriminator for prehospital triage in suspected ACS patient is based upon different studies, predominantly non-randomized trials. In such an approach, the method of selection of included studies is truly important. In this perspective, I was wondering whether both reviewers were independently from each other selecting the included studied. In other words, was the search performed twice or only 1 time, and then checked by the second observer?
3. The same holds true for the data extraction and statistical analyses of the included studies. Was this done by 1 person? And rechecked by another?
4. The study inclusion is described in Figure 1, in which initially ~ 4000 studies were identified, and at the end only 15 studies were enrolled for the systematic review. The authors mention that all articles were screened by the primary author, and if suitable the were included. What is meant with this? Please describe what criteria were used for selection and check suitability. Additionally, can the authors explain in more detail what ‘’the removal of records meant’” as well as the ‘’exclusion of records’ in step 2 of this process?
5. What was the main exclusion criterium of all these studies? Please specify the exclusion reasons

We thank the reviewer for these critical questions. We fully agree that the method of selection and data extraction is very important in this systematic review and meta-analysis. With our explanation and modifications, we hope to clarify this.

First we performed (J.D. and P.J.V.) our search in MEDLINE and a total of 1584 articles were found. After this, we searched EMBASE and a total of 2341 articles were found. EMBASE also searches in MEDLINE, so many articles we already found in MEDLINE were also in the EMBASE search. We

added a filter to the EMBASE search to exclude MEDLINE articles and therefore 2120 articles were removed before screening. After all, these had already been found in MEDLINE. The remaining 1805 articles have been screened for possible suitability by the first author (J.D.). At this first screening of the titles and abstracts, a total of 1687 articles could already be excluded. This first screening was performed based on the following 4 inclusion criteria: 1. Original data; 2. Patients presenting with NSTE-ACS; 3. Prehospital risk assessment or triage; 4. Published in English, French or German. If the article did not meet these inclusion criteria, it was excluded. The full text of the remaining 116 articles were reviewed in detail by J.D. and P.J.V separately. and checked whether they met our inclusion and exclusion criteria as described in the methods section. Any disagreements between these 2 reviewers were resolved by the decision of a third independent reviewer.
The data extraction was performed independently by two authors (J.D. and P.J.V.) and reached consensus in case of discrepancies. The statistical analysis was performed separately by two authors (J.D. and J.Z.) and checked by a third author (M.V.).

We added/modified the next sentences in our methods section in the manuscript:

“Disagreements between reviewers were resolved by the decision of a third independent reviewer.” (Page 7)

“ For included studies individual data of sample size, number of true positive, true negative, false positive and false negative were extracted separately by two authors (J.D. and P.J.V.). When possible, we excluded the STEMI patients from the data extraction. Sensitivity, specificity, likelihood ratios (LRs), and diagnostics odds ratio (dOR) were calculated from the extracted data by two authors (J.D. and J.Z.). The statistical analysis was checked by a third author (M.V.).” (Page 8)

“Double records removed” (Page 6, Supplemental material)

5. There is in EMS studies always heterogeneity, what can influence the results of the systematic review. How did you address these study heterogeneity, can you explain in more detail? Especially, the study included both GP and EMS patient populations (patient heterogeneity) as well as different study endpoints (some had 3-day MACE rate, while others had final hospital diagnosis such as ACS or NSTEM/UAP). The authors should discuss the potential effects of heterogeneity in enrolled studies on their findings

We fully agree with the reviewer regarding the heterogeneity in EMS studies. We tried to address this by calculating the I2 for characteristics evaluated in 3 studies or more. This is mentioned on page 9 in our manuscript.

“Statistical heterogeneity was reported as the I2 for the sensitivity and specificity when findings were evaluated in 3 studies or more. We considered values between 0% and 30% as low heterogeneity, between 30% and 60% as moderate heterogeneity and 60% or more as considerable heterogeneity.20”
Our review includes both primary care setting and EMS patients. To minimize heterogeneity, we calculated the diagnostic test accuracy of all the characteristics and tools separately for these two settings. No data is combined between these two settings. The setting in which each characteristic or diagnostic tool is studied is listed in all tables (Table 1 – 6 in manuscript; eTable 1 - 4 in Supplemental material). It is true that most studies had different endpoints, leading to a considerable amount of heterogeneity, especially in risk factors and POC-Troponin. This is described under the limitations section in the discussion.
“In the analysis of some variables, heterogeneity was considerable in our meta-analysis (e.g., cardiovascular risk factors and POC-Troponin measurement in the EMS setting). In POC-Troponin,

heterogeneity is the result of variation in thresholds between studies. Although this is expected in systematic reviews of diagnostic test accuracy, it is possible this influences our main findings.46 However, given the small contribution we do not think that the heterogeneity in our meta-analysis alternates the influence of stand-alone risk factors in early risk stratification for ACS." (Page 26-27)

6. Furthermore, why were studies with specific patient populations and small studies with less than 100 samples excluded? Some prehospital parameters can in such way be biased; for instance in STEMI patients, other ECG parameters are probably more relevant in STEMI patients than in patients with NSTEMI or thoracic chest pain without troponin release.

The primary objective of our systematic review and meta-analysis was to review and compare available diagnostic tools to investigate which tool has the best performance for prehospital risk assessment in patients suspected for NSTE-ACS. Currently, no prehospital diagnostic tools are used for risk assessment in this population. Only an ECG is made to distinguish between a patient with a ST-segment elevation myocardial infarction (STEMI) and a patient suspected of NSTE-ACS. For the STEMI population, there is already an excellent working prehospital triage and fast access pathway (direct transfer to PCI center) based on the ECG to minimize the time from first medical contact until revascularization. Therefore, additional diagnostic tools (e.g. POC-Troponin and Combined Risk Scores) are not of added value for this population in the prehospital setting and we excluded all STEMI patients. Because patients with undifferentiated chest pain suspected for NSTE-ACS is a large and varied population, we excluded specific patient populations and small studies.

7. Can the authors be more detailed about the extracted data regarding ECG parameters (An "ischemic ECG" is rather a broad description, what is meant with this – ST depression, Negative T waves, ST segment elevation) and biomarkers (which specific generation or essay was used) ?

A total of 3 studies investigated the value of the ECG. Unfortunately, only 1 study mentioned the definition of an ischemic ECG: ST-deviation ≥800 µV. The other studies were not clear about their definition of an ischemic ECG. This is mentioned on page 12 and in Table 1 on page 13.
The included studies used different biomarkers with different assays. The manufacturer, assay, and positive threshold are mentioned in eTable 5, page 13 of the supplemental material.

8. One of the limitation of EMS studies is often the lack of well-described pre-hospital patient and ECG data. For this reason, the lack of power of these prehospital parameters may be partly explained by lack of description. This should be discussed in more detail in the paper.

We fully agree with this comment and suggestion of the reviewer. Therefore, we added the next paragraph to the discussion.

"Another limitation is the lack of well-described patient and additional diagnostic tools data. For example, only 1 of 3 studies described their definition of an ischemic ECG. Therefore, it is difficult to draw conclusions about the diagnostic value of the ECG in this particular population in the prehospital setting. On the other hand, these limited available data is not uncommon for these kind of studies. It is the result of the acute setting in which these patients presents themselves to the GP or the EMS." (Page 27)

9. The authors found that combined risk scores and biomarkers were better predictors in the prehospital scene than conventional patient, physical examination or ECG parameters. Any explanation for this finding (is the sum of parameters more reliable, than a single-parameter)?

Our idea is that the sum of the parameters is more reliable than a single parameter. Especially, the addition of the POC-Troponin to the combination of the history, risk factors, and ECG is of added

value. Therefore, the combined risk scores have the best performance in making an NST-ACS more or less likely.

10. The value of combined risk scores depends highly on the medical experience, setting and sort of score you used. In general, the (pre) HEART scores are more often used at emergency departments or EMS, whereas the Marburg score is used in primary care setting. One should be cautious about drawing conclusions of the value of these combined scores in a particular setting.

We also agree with the reviewer on this point. Therefore, in our systematic review and meta-analysis, we used only the data from the combined risk scores for the specific prehospital setting in which they were studied. For example the (pre)HEART scores for the EMS setting and the Marburg Heart Score for the primary care setting. To make this clearer, we have modified the following sentence in the conclusion.

“On-site assessment of biomarkers and combined risk scores derived from the HEART score showed to be strong predictors for diagnosing patients with a NSTE-ACS and to identify patients without a NSTE-ACS in the EMS setting.” (Page 28)

Responses to reviewer #3

This is a well thought of and undertaken systemic review on an important topic, that of identifying patients with ACS presenting in the prehospital or primary care setting.

We thank the reviewer for his comments and compliment.

1. My main concern with this paper is that the review deals with only NSTE-ACS and this needs to be stated more clearly in the text throughout the manuscript. Also, emphasis needs to be made on the rather limited group of patients that constitutes the NSTE-ACS group. This can be further elaborated on in the discussion. When constructing risk stratification models it is a shortcoming that a part of the patient population, in this case the STEMI are excluded on beforehand. This could be even further stressed in study limitations.

We thank the reviewer for sharing these concerns. We agree that throughout the manuscript we were not clear enough that we were only interested in patients suspected of NSTE-ACS. We have changed the following sentences to make this clearer.
“Prehospital risk assessment in patients suspected of Non-ST-segment elevation Acute Coronary Syndrome – A systematic review and meta-analysis” (Title, page 1)

“To review, inventory and compare available diagnostic tools and investigate which tool has the best performance for prehospital risk assessment in patients suspected of non-ST-segment elevation Acute Coronary Syndrome (NSTE-ACS).” (Page 2)
“Sensitivity, specificity, and, likelihood ratio of findings for risk stratification in patients suspected of NSTE-ACS.” (Page 2)
“We performed a systematic review and meta-analysis of available prospective data to summarize and compare current available and new diagnostic tools for prehospital risk assessment in patients suspected of NSTE-ACS.” (Page 6)
“As the focus of this review was to investigate which patient characteristics and diagnostic tools were used for early risk assessment in an undifferentiated prehospital population suspected of NSTE-ACS..” (Page 7)
Currently, the ECG is the only risk stratification and tool used in the prehospital setting. The ECG distinguishes between patients with a STEMI and patients suspected of NSTE-ACS. When a STEMI

is diagnosed on the ECG, these patients are immediately transferred to a PCI center. This prehospital triage reduces the total ischemic time and has improved clinical outcome in this population (Keeley et al, 2003). So in this patient population, an excellent working risk stratification model already exists and therefore we excluded these patients. On the other hand, patients suspected of NSTE-ACS undergo no further prehospital risk assessment and triage and are transferred to the nearest hospital (with or without PCI facilities). Although the latest ESC guideline for the management of patients with NSTE-ACS recommends invasive coronary angiography and a transfer to a PCI-center within 24 hours. (ref 6) Most hospitals cannot follow this guideline recommendation. Therefore, we wanted to investigate which diagnostic tool could be used prehospital to help making triage decisions in this patient population specifically. Future research needs to make clear if these diagnostic tools could determine a different triage and/or treatment strategy. (Page 26)

VERSION 2 – REVIEW

REVIEWER	Boogers, Mark Leiden Universitair Medisch Centrum, Cardiology
REVIEW RETURNED	10-Feb-2022

GENERAL COMMENTS	The study addresses an important and timely topic. The authors have improved their study in a significant manner, predominantly the interpretation and clinical meaning of study findings has been clarified. Probably, the amount of information in all tables can be presented in a more compact way.